The Urban Heat Island effect (UHI) is a phenomenon in which the temperature in urban areas is hotter than in surrounding areas. With the rapid urbanization, climate changes, and aging patterns seen around the world, the threats of the urban heat island effect on environmental justice and urban health will increase. The negative effects of UHI might increase with rapid expansion of build-up areas, loss of green spaces, and global and regional heating driven by human activities, making the residents of fast-growing cities more vulnerable.

As a measure for the negative effects of urban heat islands, increasing the number of green areas is important. Greenery in and around cities improves urban microclimate by providing shading and evapotranspiration, which effectively cools surrounding areas. However, the distribution of the green spaces and their cooling benefits is not always equal across different social groups and different geographic contexts. That’s why prioritizing urban areas for urgent reforestation to guide limited resources effectively is essential for urban policymakers.

Many previous researches, including those of Li et al. (2026) and Elmarakby et al. (2024), suggested using IPCC (Intergovernmental Panel on Climate Change) guidelines with risk and accessibility variables. In that case, it’s essential to answer two questions: how much an area is vulnerable and exposed to heat and how much this area can respond to extreme weather events through access to cooling facilities. Sohn et al. (2026), later, strengthen the role of education, healthcare, and transportation in the adaptive capacity of residents in risk-exposed regions.

Despite these efforts made globally, Central Asia remains one of the most understudied regions in the world in the analysis of environmental justice, particularly green equity. Particularly in capital cities like Astana, Tashkent, and Bishkek, which had seen significant increases in the number of residents, rapid urbanization, and fragmentation of greenery, comprehensive studies in risk assessment are not available yet despite the increasing demand for thorough research on the intersection of public health and urban policy.

The purpose of this paper is to analyze urban equity in Tashkent through the use of IPCC guidelines, using vegetation, population, and facility data acquired through satellites and open-source platforms. The city expanded rapidly over the few decades, experiencing rapid growth in impervious surface and loss of green spaces, such as trees, shrubs, and grasslands (Alikhanov et al., 2020).

Specifically, we replicated the H-E-V (hazard-exposure-vulnerability) model in the context of Tashkent. Next, we integrated the adaptive capacity as the accessibility to cooling and healthcare facilities. Our study has also focused on the general green equity in the city through the implementation of the Gini coefficient. Finally, combining all results above, we constructed a priority-indexed map of Tashkent with areas falling into different urgency categories.